

Battery Build-out and the Shape of California Electricity Prices

A conditional-association study of the SP15 day-ahead evening-minus-midday spread, May 2023-December 2025

1. Introduction

California's rapid battery build-out is intended partly to shift electricity from solar-rich hours into the evening, when solar output falls and demand often remains high. This paper asks whether greater operating utility-scale battery power capacity was associated with a narrower day-ahead price spread at the CAISO SP15 trading hub. The daily outcome is the average 17:00-20:59 locational marginal price (LMP) minus the average 10:00-15:59 LMP. The sample covers 1 May 2023 to 31 December 2025.

The fully adjusted estimate is $\hat{\beta} = 0.645$ USD/MWh per GW. Its seven-lag Newey-West standard error is $SE_{HAC} = 8.012$ and its 95% confidence interval is $[-15.059, 16.349]$. The estimate is therefore highly imprecise: the data are consistent with a material narrowing, almost no conditional association, or a material widening of the spread. This is not evidence that batteries have no economic role. Rather, battery capacity and the time trend move almost together, making their separate associations difficult to identify in this short observational sample.

The paper's contribution is deliberately narrow. It does not optimise battery dispatch or estimate a policy counterfactual. Instead, it asks whether the observed build-out contains enough independent variation to support a precise conditional association after economically important controls are included. This focus follows the course requirement to organise a small number of well-motivated results around one coherent econometric argument.

2. Economic mechanism and focused literature

The mechanism begins with California's daily net-load pattern. Abundant solar generation tends to depress net load and prices around midday. In the evening, solar output falls while consumption may remain high, increasing the need for other supply. A battery can charge during the low-price period, adding demand and tending to raise the midday price; it can discharge later, adding supply and tending to lower the evening price. Both movements predict a smaller evening-minus-midday spread as usable storage capacity expands. CAISO's market-monitoring reports document the growing importance and

characteristic daily charging and discharging pattern of batteries in the California market (California ISO Department of Market Monitoring, 2025).

This reasoning is consistent with electricity-storage arbitrage: storage buys energy in low-price hours and sells in high-price hours. Sioshansi et al. (2009) show that arbitrage value depends on prices, efficiency, capacity, fuel mix and transmission conditions, and that storage itself can alter prices when deployed at scale. The spread used here is therefore an intertemporal price signal, not realised battery profit. It excludes round-trip losses, degradation, operating costs, quantities dispatched and ancillary-service revenue. The ex ante prediction is a negative battery coefficient, but profitable expected spreads may also encourage entry. That reverse association is why the design is explicitly associational rather than causal.

3. Data and variable construction

The dataset contains 976 possible Pacific-time calendar days. One day, 2 November 2024, is removed because the official adjusted EIA-930 demand and solar fields are blank for all 24 hours; 975 complete days remain. SP15 day-ahead total LMPs come from CAISO OASIS. A day is retained only if all four evening and all six midday prices are present, so the outcome has a fixed hourly composition. Source timestamps are preserved in UTC, converted once to America/Los_Angeles, and merged by local operating date. Genuine negative prices and scarcity spikes are retained rather than winsorised.

Operating battery power capacity is constructed from EIA-860M units in the CAISO balancing authority and measured in GW. Each day in a month receives the previous month-end inventory, preventing look-ahead and approximating capacity available at the month's start. The controls are the evening-minus-midday CAISO demand gap, the corresponding solar-output drop (both in GW), and CAISO's daily FRSCE3 Southern California delivered natural-gas-cost proxy (USD/MMBtu). Day-of-week indicators, month indicators and a linear trend absorb recurring calendar patterns and gradual change. Capacity varies mainly across 32 monthly levels, so 975 daily rows do not provide 975 independent changes in the exposure.

The demand and solar controls use the same hourly windows as the dependent variable. This matters because a large evening demand increase or solar decline can widen the spread mechanically even if battery capacity is unchanged. The total LMP is used rather than only its energy, congestion or loss component because it is the market price relevant to the arbitrage signal. Avoiding the 02:00 transition hour reduces daylight-saving ambiguity, while the explicit UTC-to-Pacific conversion prevents an hour from being assigned to the wrong operating day.

4. Econometric method and evidence standard

The primary specification is an ordinary least squares conditional-mean regression:

$$\begin{aligned} \text{Spread}_t = & \alpha + \beta \text{Battery}_t + \gamma_1 \text{DemandGap}_t \\ & + \gamma_2 \text{SolarDrop}_t + \gamma_3 \text{Gas}_t + \sum_{d=2}^7 \delta_d D_{dt} \\ & + \sum_{m=2}^{12} \mu_m M_{mt} + \lambda \text{Trend}_t + u_t \end{aligned}$$

Here β is the expected USD/MWh difference in the spread associated with 1 GW more battery capacity, holding the included variables fixed. The indicators D_{dt} and M_{mt} represent day-of-week and month effects; Monday and January are the omitted categories. Three nested models were estimated: capacity alone; capacity plus the three market fundamentals; and the full specification above. The full specification is the baseline because the question concerns the battery association after adjustment for fundamentals, seasonality and trend. The spread is kept in levels because it can be zero or negative, making its natural logarithm undefined.

For β to have a conditional-mean interpretation, the remaining disturbance must satisfy $E(u_t | X_t) = 0$, where X_t denotes the included regressors. This requires correct timing and measurement and no important omitted influence that is correlated with capacity after conditioning. The assumption is demanding here because storage investment is chosen rather than randomly assigned. Month effects remove recurring seasonality and the trend absorbs a smooth common movement, but neither guarantees that changing policy, generation composition or network conditions are controlled. Consequently, OLS is treated as a best linear description of the observed joint variation.

Daily regression errors can be heteroskedastic and serially correlated. The reported inference therefore uses Newey-West heteroskedasticity-and-autocorrelation-consistent standard errors with seven lags. The two-sided null is $H_0: \beta = 0$; the pre-specified economic prediction $\beta < 0$ is discussed separately to avoid choosing a one-sided test after observing the estimate. Residual-versus-fitted and residual-over-time plots, a 30-day residual autocorrelation function, the Durbin-Watson statistic and a seven-lag Breusch-Godfrey test were produced. Correlations and variance-inflation factors (VIFs) assess multicollinearity. Supporting checks include augmented Dickey-Fuller tests, influence diagnostics, a monthly first-difference model, component-price regressions and alternative capacity, timing and price outcomes. These diagnostics can change the credibility or precision of inference; they cannot turn deployment into an exogenous treatment.

5. Results and diagnostics

Table 1. Battery-capacity association with the SP15 price spread

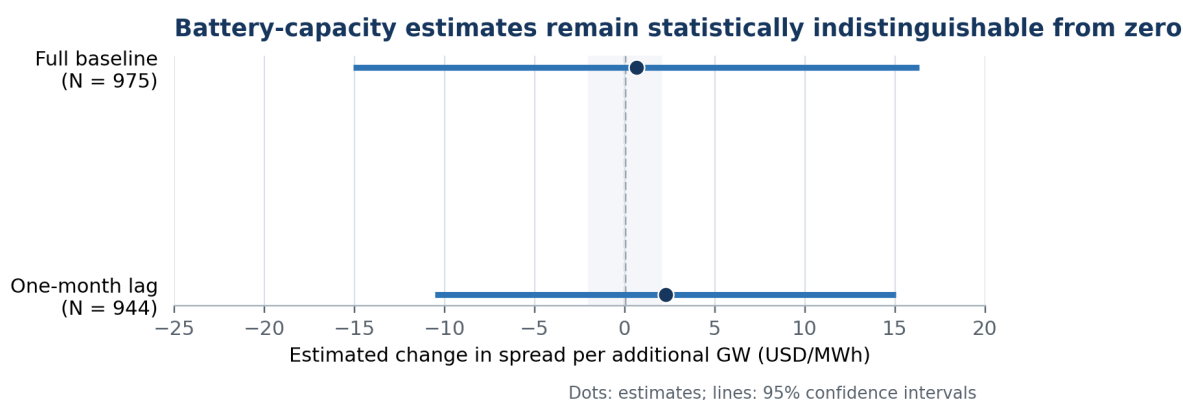
Specification	Coef.	HAC SE	95% CI low	95% CI high	p-value	N
Full baseline	0.645	8.012	-15.059	16.349	0.936	975
Capacity lagged one additional month	2.271	6.524	-10.516	15.058	0.728	944

Notes: The dependent variable is the daily average 17:00-20:59 SP15 day-ahead LMP minus the average 10:00-15:59 LMP, in USD/MWh. Both rows use the full controls, day-of-week and month effects, a linear trend, and Newey-West standard errors with seven lags. The lag-model confidence interval is calculated from the reported estimate and standard error.

The baseline point estimate is small relative to its uncertainty. The test statistic is $t = \frac{0.645}{8.012} \approx 0.08$, and $p = 0.936$ provides virtually no evidence against a zero conditional association. More importantly, the wide confidence interval contains economically different possibilities. At its lower end, an additional GW is associated with a spread about 15.06 USD/MWh narrower; at its upper end, the spread is about 16.35 USD/MWh wider. It would therefore be incorrect to interpret the positive point estimate as evidence that batteries widen spreads, just as it would be incorrect to claim that the relationship is proven to be zero.

The interval also shows why statistical and economic significance must be separated. The central estimate is economically small, but the uncertainty range is not: it includes changes large enough to matter for the intertemporal price signal. A non-significant test therefore reflects weak precision, not a tightly estimated zero. The result answers the research question by limiting what can be concluded from these data, even though it does not confirm the predicted negative sign.

Figure 1. Point estimates and 95% confidence intervals



Notes: The vertical dashed line marks zero. The one-month-lag interval is approximate, using $1.96 \times SE_{HAC}$.

The main reason for this imprecision is multicollinearity. Battery capacity and the linear trend have a correlation of 0.9975, while their VIFs are approximately 390. California's battery fleet expanded strongly over the same period represented by the trend. The regression must therefore decide whether gradual price changes belong to battery capacity or to other time-related developments; the data

provide little independent variation with which to do so. Multicollinearity does not automatically bias OLS, but it inflates the sampling variance and makes the battery coefficient unstable. Removing the trend simply to obtain a smaller standard error would change the identifying assumption and risk attributing all gradual market change to batteries.

This distinction is important for model choice. A capacity-only regression would use the common upward movement in capacity and other market conditions as if it were battery-specific variation. The full model asks a harder but more relevant question: whether days associated with different monthly capacity levels have different spreads after fundamentals, calendar patterns and a general time path are held fixed. The resulting loss of precision is a substantive feature of the data, not a reason to discard a pre-specified control after seeing the result.

The timing check reinforces rather than resolves this conclusion. Lagging the already month-start capacity series by one additional month reduces the sample and yields $\hat{\beta}_{lag}=2.271$, $SE_{HAC}=6.524$, $p=0.728$, and $N=944$. Its interval also spans sizeable negative and positive values. The residual plots, autocorrelation diagnostics and conventional-versus-HAC comparison were retained in the reproducibility output; $HAC(7)$ remains the reported inference because price-series errors can be dependent even when a single diagnostic is reassuring. The pre-specified energy-capacity, component-price, tariff-break, outlier, first-difference and negative-price-hour checks likewise did not produce evidence precise enough to overturn the baseline conclusion. This selectivity follows the course guidance: these checks test the robustness of the central interpretation rather than create a search for statistical significance.

6. Interpretation and limitations

The result is a conditional historical association, not a causal storage effect. First, reverse association is plausible: developers may enter because they expect persistent arbitrage opportunities. Second, battery capacity co-moves with omitted changes including solar additions, thermal retirements, transmission upgrades, hydrology, imports, outages, carbon prices, market rules and policy. Calendar controls and a trend cannot guarantee that these influences are removed. The near-perfect capacity-trend correlation is direct evidence that this identification problem is severe rather than hypothetical.

Third, nameplate capacity is not battery operation. Installed MW does not reveal state of charge, duration, degradation, availability, bids, dispatch or participation in ancillary-service markets. Fourth, the EIA inventory is preliminary, omits sub-1-MW and behind-the-meter systems, and measures utility-scale units across the CAISO balancing authority, whereas the price outcome is the southern SP15 hub. CAISO-wide demand and solar controls are also imperfect geographic matches. Finally, daily observations overstate the effective exposure variation because capacity changes monthly. The study is

consequently informative about whether the observed build-out can be separated statistically from contemporaneous market evolution; it is not a forecast of what a new battery mandate would do to prices. A stronger causal design would require a credible source of exogenous deployment timing or more granular operational variation, not simply additional controls.

Even a precisely negative coefficient would need cautious policy interpretation. A narrower spread may reflect beneficial load shifting, but the spread is not total welfare: consumers and producers can be affected differently, and reliability, emissions, congestion and ancillary services are outside the outcome. Conversely, failure to estimate a precise spread association does not imply that batteries lack value in those other dimensions.

7. Conclusion

After controlling for matched-hour demand and solar patterns, delivered gas costs, calendar effects and a trend, 1 GW more battery capacity is associated with a 0.645 USD/MWh wider SP15 evening-minus-midday spread. The estimate is extremely imprecise: $p=0.936$ and 95 % CI=[− 15.059, 16.349]. The one-month-lag result is similarly inconclusive. The correct conclusion is therefore not that batteries have no influence, but that this sample cannot distinguish their independent association from zero or from other changes occurring over time. Severe capacity-trend collinearity and non-random deployment are the central limits.

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